Saqib Bedar

Islamabad, Pakistan (Open to Remote · UTC+5)
+92-3053721981 | chandio.saqib.m@gmail.com | linkedin.com/in/saqibbedar | github.com/saqibbedar
saqibbedar.github.io

Professional Summary

Python and full-stack engineer (3+ years in web) scaling into ML application development: secure FastAPI/MERN backends and PyTorch vision-language research. Self-directed, async-first and documentation-led, working remotely from UTC+5.

Technical Skills

Languages: Python, JavaScript, C++
Backend Development: FastAPI, Pydantic, REST API Design, Node.js, Express, JWT, Role-Based Access Control
Data & Machine Learning: MongoDB, MySQL, PyTorch, NumPy, Pandas, Computer Vision, Transfer Learning
Tools & Testing: uv, Ruff, pytest, Git, GitHub Actions, GitLab, Linux, Docker
Spoken Languages: English (Professional Proficiency), Urdu (Native), Sindhi (Native)

Projects & Research

UV-JEPA (Final-Year Research, WIP) | *PyTorch, ViT-Small, RoBERTa, JEPA* 2026 – Present
- Building a latent-predictive (JEPA) vision-language model for Urdu, a low-resource language, under data-constrained training on a 59K-image / 319K-caption corpus.
- Implementing dual encoders (ViT-Small, RoBERTa) with an EMA-updated target text encoder and a cross-attention predictor optimized via normalized latent regression, avoiding autoregressive decoding errors.

MERN Student Portal Monorepo | *Node.js, Express, React, MongoDB* Dec 2025
- Architected a 4-role RBAC university portal (Master Admin, Department Admin, Faculty, Student) on a modular Model-View-Controller backend, delivered as production client work.
- Implemented server-side HTTP-only cookie authentication with JWT refresh rotation and a FIFO concurrent-session eviction algorithm capped at 3 sessions per user.
- Built rate-limited email OTP recovery (3 per day, 2-minute cooldown) and hard-delete operations that run referential-integrity validation before removal.

Fluent-Python: CI/CD Testing Ground | *Python 3.12+, uv, Ruff, pytest* 2025 – Present
- Run a production-style pipeline testbed demonstrating branch protection, local automation scripts, and multi-stage CI/CD workflows on protected branches.
- Maintain an open-source Python engineering curriculum (language internals, OOP, async, LLM integrations) with uv environments, Ruff linting, and automated pytest suites.

Date-Fruit Classification | *PyTorch, Custom CNN, ResNet-50* 2024
- Designed a custom 4-block PyTorch CNN and benchmarked it against a ResNet-50 transfer-learning backbone on 7,000 images across 10 classes, reaching 96.38% test accuracy; deployed as a real-time PyTorch inference web app.

Experience

Teaching Assistant, Web Application Development Sep 2024 – Present
Quaid-i-Azam University *Islamabad, Pakistan*
- Instructing 35+ students across practical laboratory sessions focused on Git version control workflows, REST API consumption, and OpenAI/Gemini SDK integrations.

Front-End Developer Intern 2025
College of Army Education (CAE) *8-Week Internship*
- Delivered responsive production interfaces within an 8-week cycle using a modern JavaScript/React stack.

Education

Quaid-i-Azam University Islamabad, Pakistan
BS in Computer Science *2023 – June 2027 (Expected)*

Community & Writing

Speaker: MLSA QAU (2024), AutoMind Community (2026) | Author: Gitcraft, 33-page Git documentation site